

Commercial aviation

Official predictions about the scale of future passenger air travel have often seemed excessively bullish, yet have so far proved quite accurate. In 2000, for example, the aircraft industry was predicting that 4 billion passengers a year would be carried by 2020. When in the following year the terrorist attacks on September 11 (see page 410) triggered a shocking fall in passenger numbers, these upbeat forecasts for future growth were left looking plain wrong. But the downturn proved short-lived and by 2016 annual passenger numbers had already reached 3.6 billion, with growth continuing at over 4 percent a year. Based on the previous accuracy of forecasts, therefore, it seemed reasonable to accept predictions for 7 billion air journeys being made annually worldwide by 2034. By that time there might be around 40,000 commercial aircraft in operation, almost double the number flying in 2016.

Perhaps the most certain guarantee of future growth in passenger air travel was provided by the untapped potential of populous Asian countries, especially China, to become a major new source of air passenger business. In 2000 China had around 527 airliners carrying some 67 million passengers a year. By 2015 the comparable figures were 2,500 airliners carrying 438 million passengers annually. China had become second only to the United States in the scale of its passenger operations and,

TRANSPORTED IN SECTIONS

A convoy carries sections of an A380, made in Britain, France, Germany, and Spain, from a barge terminal on the Garonne river to the Airbus assembly plant at Toulouse, France. Crawling along rural backroads, the massive convoy takes three nights to complete its trip.

ENGINE TESTING

The Airbus A380's massive Rolls-Royce engines underwent rigorous testing to ensure that they met targets for fuel consumption, emissions, and noise, as also safety. One requirement for air safety certification was that the engine would still function if a large goose flew into the fan.

with expansion continuing at over 10 percent a year, was set to overtake America sometime in the 2030s.

Airbus versus Boeing

If air travel continues its spectacular expansion, what aircraft will future passengers be boarding? In the first 30 years of the jet airline era, Boeing had established a seemingly unshakeable worldwide supremacy in this sector, but the spectacular rise of the European consortium Airbus Industrie shattered Boeing's grip on the market. In 2004 Airbus knocked the mighty Boeing off the top of the international airliner sales and orders league for the first time.

Designed for the market

The boldest Airbus project to date has been the A380 superjumbo, which had its maiden flight in April 2005. As the largest airliner ever built, capable of



carrying at least 550 passengers, it aimed to replicate the revolutionary impact the Boeing 747 had had on air travel three decades earlier. The project showed how far Europe had advanced since the days of the Concorde project – not in technological know-how, which was never lacking, but in commercial realism. The A380 was crafted to fulfil the business objectives of its potential airline customers, offering a projected 20 percent cut in operating cost per seat compared with the Boeing 747-400. Careful thought was given to minimizing the new demands these enormous aircraft would make on airports, virtually ensuring that the superjumbo would be welcome at all major air transportation hubs. The design of the aircraft also reflected the need to meet environmental

